DOES HYPERBOLIC GEOMETRY HELP GRAPH-BASED SENTENCE POOLING? A CONTROLLED STAGE-WISE STUDY

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ABSTRACT

Hyperbolic space represents tree-like structure with less distortion than flat space, and transformer token states are often described as tree-like. That pairing has motivated hyperbolic versions of many models. We ask whether it actually helps, on one architecture, under controlled conditions.

Our subject is GLOT (Mantri et al., 2026), a sentence pooler that builds a graph over the tokens of a frozen language model, refines it with a GNN, and aggregates it into a single vector. Geometry enters in exactly three places — how the graph is built (**Stage A**), how states are pooled (**Stage B**), how messages combine (**Stage C**) — so we replace each independently and give all eight combinations the same search budget over fifteen shared seeds.

The three stages disagree, and that is the main result. Stage B, the hyperbolic readout, is reliably harmful — on two encoders and a decoder, with the effect size replicating to within 0.2 Spearman across backbones — and we can say why: it amplifies seed variance rather than acting as a useful prior. Stage C, hyperbolic message passing, is the only stage that ever reaches significance, and it helps. Stage A, the hyperbolic graph, is nominally positive more often than not but never separates from the baseline. What does *not* transfer is how the stages combine: the sign of the B$\times$C interaction changes with task and backbone.

We are careful about the one gain, and we have had to become more careful. Stage C's $+1.42$ MCC on CoLA is significant end to end, but a factorial that separates curvature from the graph density and architecture it was tuned alongside cannot attribute it to either. Worse, we found that the released training code does not pin a run to its seed: re-running the identical configuration at the identical seed moves CoLA by up to 3.6 MCC, which is larger than the effect. Re-measured inside a single campaign, the curvature component of Stage C's gain falls from $+0.40$ to $+0.07$.

We also question the premise. The usual justification for going hyperbolic is a low measured δ-hyperbolicity. That statistic is an artifact of a few very large token norms: on an outlier-free metric the backbones it is used to tell apart are indistinguishable (0.2127 against 0.2114), while the Euclidean version differs between them by two orders of magnitude. Measured tree-likeness cannot tell you where curvature will help — only a controlled, component-wise comparison can.

DRAFT STATUS

Every number in this draft was recomputed from the released campaign CSVs on 30 August; nothing is transcribed from an earlier revision. The audit changed fourteen values and one conclusion, and those changes are recorded in Appendix O rather than made silently.

What is missing

Experiment	What it would settle	Status
Stage C factorial, all 4 cells × 65 seeds	Replaces Tables 6 and 7 at 80% power on the interaction, *and* removes the cross-campaign splice that Appendix D shows is unsafe	running; 65/65, 28/65, 15/65 and 0/65 of the four cells are done; ≈7 h left
Stage A on CoLA, 50 seeds	Settles the one stage the paper openly leaves unresolved	queued, ≈2.5 h
Decoder STS-B, arms B and C *alone*	The **??** cells of Table 4. Without them the paper claims a decoder replication of Stage B without ever having run B by itself on a decoder	queued, ≈11 h
Decoder CoLA, 8 arms	A second decoder task, and the first test of Stage C outside BERT and RoBERTa. Tuning rows exist; *no* confirmation seed has run	collaborator machine, ≈30 h
MTEB with a pooler actually trained on MS MARCO	Replaces the untrained-pooler block of Table 12, which currently measures only random initialisation	pipeline rebuilt and validated; not yet launched, ≈5 h
SST-2 reduced design (baseline, B, C, BC; 15 seeds)	The small-data confound in Appendix P — the one remaining way Stage B's negative could be an artifact	*not* queued: no script, and the cache must be built first
Mean / Max / `[CLS]` / AdaPool on GLUE	Would let Table 4 reproduce the *content* of the original's Table 1 rather than only its layout	reachable since the pooler was made configurable; not run
Stage C self-loop fix, 2 cells × 15 seeds	Removes the duplicate-self-loop asymmetry of Appendix N before the 65-seed job measures it at higher precision	*not* queued, ≈1.5 h
IMDB, arms baseline, A, C, AC, 5 seeds	A long-text test of Stages A and C. At $n{=}5$ this is a lead, not a result	*not* queued; cached states are 79 GB
Related work	§5	drafted; venues and pages unconfirmed for the entries marked so in `references.bib`

What could still change a conclusion

(i) The 65-seed factorial. If it attributes Stage C's gain to configuration rather than curvature, the single positive result stops being a result about geometry at all. The partial data already leans that way: recomputed inside one campaign, the curvature component is $+0.07$, not $+0.40$ (§4.3). (ii) SST-2. If Stage B's harm disappears there, the Stage B finding is a small-data artifact rather than a property of the readout — and that run is *not* scheduled, so the confound stands. Every claim below is stated at the strength today's data supports, and both of these will have to be restated in whichever direction the runs go.

1 Introduction

Turning a language model's token states into one vector is a first step for most sentence tasks. Simple poolers — mean, max, `[CLS]` — treat tokens as an unordered bag. A newer idea is to treat pooling as *relations first, aggregation second*: build a graph over tokens, refine it with a GNN, then aggregate. GLOT (Mantri et al., 2026) is a strong example, matching competitive GLUE and MTEB numbers with 20× fewer trained parameters and a frozen backbone. Our runs support the claim it rests on: on the authors' own noise test, GLOT beats every bag-of-tokens pooler by 14–35 points at every distractor ratio, in every seed (Appendix J).

This paper asks what else that design can carry. Hyperbolic space is a natural candidate for all three of its parts, so we build a hyperbolic version of each and measure them separately. That separation is the point: two of the three move in opposite directions, so a single "hyperbolic" switch would have reported their average and found nothing.

Contributions.

1. **A stage-wise hyperbolic study** (§4). Under an equal search budget with 15 shared seeds, Stage B is significantly harmful on two encoders and on a decoder; Stage C is the only stage that reaches significance in the positive direction; Stage A trends positive but never separates from the baseline. The individual stages replicate across backbones; the interaction between them does not.
2. **A seed is not a run** (§3, Appendix D). The released code sets the Python, NumPy and Torch RNGs but never requests deterministic kernels. Re-running one configuration at one seed moves CoLA MCC by up to 3.61, with paired $\mathrm{sd} = 1.47$ — larger than the $+1.42$ effect this paper reports. This is not a caveat we inherited from elsewhere; it changed one of our own numbers by a factor of six.
3. **The usual motivation does not survive measurement** (§4.5). Low δ-hyperbolicity is driven by a handful of very large token norms. On an outlier-free metric the backbones it is used to tell apart are indistinguishable.
4. **The choices that changed our answers** (§3). A cache-warming order worth 5 MCC, a search that reverses arm rankings, a threshold that silently produces a complete graph on RoBERTa, and four defects in the released code.

2 What we changed

Let $x_1, \ldots, x_L$ be the token states of a frozen backbone. GLOT joins two tokens when their cosine passes a threshold,

$$A_{ij} = \mathbf{1}[\cos(x_i, x_j) > \tau], \tag{1}$$

runs K graph-attention layers over the result, and aggregates with a learned readout. Only the pooler trains. Geometry enters in three places, so we replace each in turn. Appendix A gives the Poincaré ball, the maps $\exp_0^c$ and $\log_0^c$, and the geodesic distance we use. Figure 1 in Appendix B shows where each stage sits.

Stage A — a Poincaré graph. Lift the tokens into the ball and join them when their geodesic distance falls below the q-quantile of all pairwise distances, $A_{ij} = \mathbf{1}[\, d_c(\exp_0^c(\tilde{x}_i), \exp_0^c(\tilde{x}_j)) < \rho_q \,]$. Using a quantile rather than a fixed radius fixes the edge count at q, so A and the cosine baseline can be compared at the same sparsity.

Stage B — an Einstein gyro-midpoint readout. Replace the weighted sum with the Einstein midpoint in Klein coordinates,

$$m_K = \frac{\sum_i a_i \gamma_{k_i} k_i}{\sum_i a_i \gamma_{k_i}}, \qquad k = \frac{2x}{1 + c\|x\|^2}, \qquad \gamma_k = (1 - c\|k\|^2)^{-1/2}, \tag{2}$$

then map back and into the tangent space for the task head. Unlike a plain mean this depends on curvature: as $c \to 0$ it becomes the arithmetic mean.

Stage C — Möbius message passing. Replace each GNN layer with a hyperbolic one that lifts to the tangent space to combine neighbours,

$$h_i' = \exp_0^c\Big(\sigma\Big(\textstyle\sum_{j \in \mathcal{N}(i)} \alpha_{ij}\, \log_0^c\big(W \otimes_c h_j\big)\Big)\Big), \tag{3}$$

with $W \otimes_c h = \exp_0^c(W \log_0^c(h))$ and α_{ij} either degree-based (HGCN) or attention (HGAT).

Curvature is searched in a scale-free way. Curvature and feature scale cannot be told apart here. The geodesic distance depends only on $\sqrt{c}\,\|x\|$, so doubling the features and quartering c leaves every distance unchanged, and searching c on its own wastes most of the grid. We instead rescale features to unit mean norm and search the *effective radius* $r_{\text{eff}} = \sqrt{c}\,\mathbb{E}\|x\|$ over $\{0.5, 1, 1.5, 2, 3\}$. This is not cosmetic: with raw BERT token norms (≈ 14.7) the map $\exp_0^c$ saturates to radius 1.0 in `float32`, real pairwise distances land in $[8.85, 11.76]$, and any fixed distance threshold below about 9 returns an empty graph (Appendix N).

We run all eight combinations of A, B and C, plus a `no graph` control that deletes the token graph but keeps the trained projection and readout, and on RoBERTa two threshold-calibration arms (Appendix H).

3 How we ran it, and the choices that changed the answer

Every arm is frozen-backbone, 2 epochs, Adam, batch 32; only the pooler trains. Tuning draws 40 configurations at one held-out seed (10 on the decoder, where a run costs four times as much). Confirmation re-runs the winner on the fixed seed set $\{1, \ldots, 15\}$, shared across arms. We report the exact two-sided sign test beside the paired t-test and call something significant only when both agree, with a Bonferroni correction across the arms tested against a common baseline within a campaign. Full protocol in Appendix C. Six choices moved our results more than the effects we were trying to measure.

A seed does not pin a run, and this is the one that hurt. The code seeds Python, NumPy and Torch, but never sets `torch.use_deterministic_algorithms` or the cuDNN determinism flags, and the scatter reductions underneath message passing are non-deterministic on CUDA. We found this by accident: the extended factorial re-ran the CoLA baseline cell under a configuration string *identical* to the one already in the main campaign, and disagreed on all 15 seeds — mean $+0.334$ MCC, sd 1.472, largest single-seed gap 3.61 (Appendix D). Two consequences. First, a table that splices a cell from one campaign into another is unsafe, and our own published decomposition of Stage C did exactly that (§4.3). Second, pairing buys less than we claimed: on STS-B it still halves the variance, but on CoLA the paired difference is *noisier* than the independent one, so the seed explains almost nothing there. All within-campaign comparisons in this paper are unaffected — both arms of every paired test were run in the same campaign, and re-run non-determinism sits inside the paired scatter the tests already estimate — but we would not have known that without checking.

The search space must be the same for every arm. An earlier campaign of ours searched only graph knobs and left the learning rate, depth and width pinned at values picked for a *flat* pooler. That handed the baseline an advantage and produced a false positive we retract in Appendix L: Stage A on STS-B looked like $+0.223$ Spearman with $15/15$ seeds and $t(14) = 6.20$, and collapsed to $+0.028$ once the baseline was allowed to pick $K = 4$. Every BERT and RoBERTa campaign here uses GLOT's full grid applied identically to all arms: 40 trials each, 15 seeds, 495 runs per task on BERT and $1{,}980$ across the four GLUE tasks. What has to be equal is the budget *within* a campaign, not the same number everywhere — the decoder gave every arm 10 trials rather than 40, for cost.

The search really does use that freedom. The learning rate does not settle on one value. On STS-B all nine arms chose 2×10^{-4}; on CoLA eight of nine chose 1×10^{-3}; MRPC and RTE split. Depth, width and threshold move too. Arms were therefore not locked to a rate chosen for a flat pooler, which is the main form of the objection that non-Euclidean layers get judged under flat-tuned optimisers. Appendix E lists the selected configurations.

Tuning maxima are inflated, and can reverse a ranking. With T trials and per-seed noise σ, the best of T sits above the truth even for an arm identical to the baseline — about 1.83 here on STS-B. On the RoBERTa campaign the inflation is large enough to flip the order outright: the quantile correction is first of five at the tuning maximum (56.77) and last of five over 15 seeds (53.19). Reporting tuning maxima would have inverted our conclusion. We report confirmation means only.

Cache order is worth about 5 MCC. The released code caches backbone states and builds the shuffled loader *before* the classifier is created. On a cache miss the loader consumes the global RNG; on a hit it does not. The same seed therefore gives a different initialisation depending on cache state — 40.37 cold against 45.54 warm on CoLA, about six times the seed standard deviation. We build all caches before any arm runs. Any reproduction that skips this will shift by several points.

The estimator matters as much as the method. Mantri et al. (2026) fix seed 42 and report the best τ per task on the split they also report, so every published number is one draw maximised over a search. We report 15-seed means. On CoLA the choice of estimator alone moves the same runs by nearly 5 MCC (Appendix F). Matching their estimator exactly, we reach 50.52 MCC on CoLA against a published 47.49 and 59.57 on RTE against 59.21; STS-B stays 0.87 short and we could not explain it. Every comparison in this paper is against our own baseline under identical conditions, never against a published number.

4 WHAT WE FOUND, STAGE BY STAGE

Table 4 gives every absolute score, laid out like Table 1 of Mantri et al. (2026); Table 5 gives the paired differences this section discusses. Both are in Appendix G.

4.1 STAGE A — THE HYPERBOLIC GRAPH: UNRESOLVED

Stage A is nominally ahead of the baseline in five of the seven settings we ran it in: $+0.70$ MCC on BERT CoLA, $+0.03$ Spearman on BERT STS-B, $+0.04$ F1 on MRPC, $+0.02$ accuracy on RTE and $+0.04$ Spearman on the decoder, against -0.23 and -0.09 on RoBERTa. It is also ahead of the baseline at all four distractor ratios of the noise diagnostic (Appendix J). But it reaches significance in none of them — the strongest is CoLA, at $t = 2.37$ with a sign-test p of 0.118, which fails our both-tests rule — and a sign test across the seven settings gives $p = 0.45$.

Verdict: unresolved, neither a positive nor a null. The direction is suggestive and the evidence is not there. A larger seed budget on CoLA is the cheapest way to settle it, and that run is queued at $n = 50$.

[TODO: When the n=50 CoLA run lands it should turn "unresolved" into either a positive or a null. Commit to whichever it returns — the run is powered in advance, so re-running it at a different n after seeing the answer would be exactly the practice this paper criticises.]

4.2 STAGE B — THE HYPERBOLIC READOUT: HARMFUL, AND THIS IS THE FIRMEST RESULT IN THE PAPER

Every arm containing Stage B loses 0.75–1.24 Spearman on BERT STS-B, with 0 or 1 of 15 seeds positive. B alone gives $t = -17.00$ with $0/15$ seeds and a sign-test p of 6×10^{-5}, the smallest value 15 paired seeds can produce. All four B-containing arms survive Bonferroni. The result repeats on a decoder backbone, TinyLlama, where AB costs 3.56 Spearman and ABC costs 4.51, both $0/15$ and both Bonferroni-significant.

It also repeats on a *second encoder*, and the agreement is close enough to be worth stating numerically. On RoBERTa STS-B, B alone costs 1.264 Spearman ($t = -15.76$, $0/15$) against 1.065 on BERT ($t = -17.00$, $0/15$), and AB costs 0.795 ($t = -5.77$, $1/14$) against 0.747 ($t = -5.66$, $1/14$). Both survive Bonferroni on both backbones. Stage B alone is negative in all four encoder settings we can measure it in, and 15 of the 16 B-containing cells across those settings are negative.

There is a mechanism, not just a mean. On the decoder the seed standard deviation rises in step with the number of hyperbolic parts: $0.49 \to 0.95 \to 1.87 \to 3.58$ for baseline, BC, AB and ABC, a $7.3\times$ spread. The gyro-midpoint behaves like a variance amplifier rather than a useful prior. It is also the stage whose theoretical motivation is weakest here: pooling averages tokens the graph has already smoothed, and the Einstein midpoint's curvature dependence pulls hardest exactly on the large-norm tokens that Appendix H shows are outliers.

One task disagrees, and we state it rather than bury it. On RTE all four B-containing arms are *positive* ($+0.79$ to $+1.85$ accuracy). None is significant, and none could be: RTE has 277 evaluation examples, a seed standard deviation of 3.36, and a minimum detectable effect of 2.06 points — larger than any effect reported in this paper. We read RTE as uninformative rather than as contrary evidence, but it does mean the honest claim is "B is harmful on the tasks that can measure it".

Verdict: harmful. The one claim here that has replicated rather than merely extended. The single remaining way it could be an artifact is scale — every task we ran has at most $8{,}551$ training examples, and variance amplification is a quantity more data suppresses. SST-2 would settle it and is not scheduled.

[TODO: B *alone* has never been run on a decoder; only AB, BC and ABC have. Do not widen "repeats on a decoder" into a claim about B in isolation there until the queued rows land, and do not quietly drop them if they disagree.]

4.3 Stage C — Hyperbolic message passing: the only stage that ever helps, and we cannot say why

Stage C reaches $+1.42$ MCC on BERT CoLA with $13/15$ seeds positive ($t = 4.11$, sign $p = 0.0074$) — significant on both tests, though just above that campaign's Bonferroni line of 0.0063. On BERT STS-B, A+C is the best of the nine arms ($+0.24$) and C is second ($+0.20$). Every arm with the hyperbolic GNN and *without* the readout is positive on both tasks.

RoBERTa agrees in direction and adds nothing in strength. C is the best arm on RoBERTa CoLA ($+0.64$, $8/7$) and the best *hyperbolic* arm on RoBERTa STS-B ($+0.29$, $10/5$), so Stage C is nominally positive in all four encoder settings — but neither RoBERTa figure is significant on either test, and a sign test over the four settings gives $p = 0.125$. Four out of four in the same direction is worth reporting; it is not worth calling a result. Stage C also does not reach MRPC, where every arm containing B or C is slightly negative (-0.15 to -0.33, none significant) and only Stage A and `no graph` are nominally ahead.

And it does not decompose. The search had given C a $3.6\times$ denser graph than the baseline ($q = 0.38$ against 0.10) together with half the depth and half the width, so the $+1.42$ mixes geometry with those choices. We ran the 2×2 that separates them (Table 6). Table 7 reports the decomposition two ways, and *the two ways disagree on the only component that matters*:

Component of Stage C's CoLA gain	spliced	within one campaign	agree?
geometry alone (curvature, at the baseline's own graph)	$+0.400$	$+0.066$	no
configuration alone (denser graph, smaller net)	-0.012	-0.346	no
interaction	$+1.028$	$+1.362$	no
total	$+1.416$	$+1.082$	—

The "spliced" column subtracts the baseline cell as it was recorded in the main campaign; the "within one campaign" column subtracts a re-run of that identical configuration inside the factorial itself. They differ only in which draw of the same cell is used, and §3 explains why those draws are not interchangeable. Neither column has a significant component: the largest t among the six is 1.66, because each component differences two independently trained cells and so carries 2–3$\times$ the variance of the total.

Verdict: real but unattributed, and less attributable than we first reported. The end-to-end effect is significant on both tests. The claim that curvature is what produces it rested on the $+0.40$ geometry term, and inside a single campaign that term is $+0.07$ with 7 of 15 seeds positive. Stage C's gain needs both the denser graph and the curved message passing; which does the work is open. That C is negative on MRPC, where the search chose a $15\times$ *sparser* graph, is consistent with needing both, but is not evidence for it.

[TODO: Written at $n = 15$ with three of four cells re-run inside the factorial and the fourth still spliced. The 65-seed run of all four cells is in flight and is powered to resolve the interaction; when it lands this subsection and Tables 6–7 get rebuilt from that file alone. If it attributes the gain to configuration rather than curvature, the paper's one positive result is no longer a result about geometry — report that outcome as readily as the other one.]

4.4 How the stages combine does not transfer

Stage B's harm is stable; what happens when the hyperbolic GNN is added on top of it is not. Adding C to B rescues the arm in two settings and worsens it in two others:

Setting	B alone	BC	effect of adding C
BERT STS-B	-1.07	-1.24	worse
RoBERTa STS-B	-1.26	-0.26	rescues
BERT CoLA	-1.26	$+0.22$	rescues
RoBERTa CoLA	-0.37	-2.16	much worse

The extremes are not noise: BC and ABC on RoBERTa CoLA lose 2.16 and 3.31 MCC and both survive Bonferroni, while the same two arms on RoBERTa STS-B are flat (-0.26, -0.13, neither significant). So a two-component summary of the form "B harms, C offsets it" is not supported. Stage B's harm is a property of the readout, reproducible across backbones and tasks; the sign of the B$\times$C interaction is a property of the particular task and backbone, and we cannot predict it.

Why the stage-wise design was necessary. B and C carry opposite signs. A study that turned "hyperbolic" on as one switch would have reported their average, which is close to zero, and concluded that geometry does not matter here. It does — the parts disagree, and only a stage-wise test can see that. The cross-backbone comparison sharpens this rather than softening it: because the size and even the sign of the interaction changes with task and backbone, the average an aggregate switch reports is not a stable quantity either. Two studies running the same all-or-nothing comparison on two backbones could report opposite conclusions without either being wrong.

4.5 The premise: measured tree-likeness is an artifact

Hyperbolic layers are usually justified by measuring Gromov δ-hyperbolicity and finding it small. On the Euclidean metric ModernBERT at layer 16 gives $\delta = 0.0021$, which reads as almost perfectly tree-like. But that layer has a maximum-to-median token norm ratio of 156, and δ is scale-sensitive: a few very large "attention sink" tokens dominate it and pull it toward zero — a star graph is a tree. On the angular metric $\arccos(\cos(\cdot,\cdot))$, which ignores those norms, ModernBERT gives 0.2127 and BERT gives 0.2114: indistinguishable, while the Euclidean numbers differ by two orders of magnitude. Flan-T5 shows the same pattern with $357\times$ norm outliers. So measured tree-likeness cannot tell you which backbone or which component will benefit from curvature. We did not use it to choose; we applied all three stages everywhere and let the paired tests decide.

4.6 Porting the method to a new backbone

Equation 1 compares a cosine to a fixed number, which only makes sense if cosine scale is comparable across models. It is not. On RoBERTa the *tenth*-percentile token cosine is 0.693, above the largest τ in the published grid, so every searched setting yields an essentially complete graph (0.991 density; Table 8). The useful finding is what happens next: GLOT is unharmed by that dense graph, while the obvious repair — switching to a quantile threshold — costs 1.41 MCC ($2/13$, $p = 0.0074$; significant uncorrected, not after the Bonferroni correction the now-larger RoBERTa campaign requires). It also won its tuning stage before finishing last of the calibration arms over 15 seeds, so selection actively favoured it. We therefore withdraw density-matching as a recommendation for anyone porting this method.

And the graph itself carries no measurable GLUE signal on either encoder. Deleting the token graph outright costs 0.29 MCC on RoBERTa CoLA ($5/10$) and 0.05 Spearman on RoBERTa STS-B ($7/8$), neither close to significant. On BERT the same control is if anything mildly *positive*: $+0.83$ MCC on CoLA ($11/4$, $p = 0.118$, second of nine arms), $+0.03$ Spearman, $+0.03$ F1 and $+0.07$ accuracy, none significant. So at every tuned density we searched, the edges contribute nothing measurable on GLUE. We deliberately do not read this as a verdict on GLOT: `no graph` is still a trained pooler with the same projection and readout, and the graph *does* beat real bag-of-tokens poolers by 14–35 points on the noise diagnostic. But it is the bar any geometry claim has to clear first, and Stage C clears it on BERT STS-B ($+0.16$, $12/3$) and fails it on BERT CoLA ($+0.58$, $8/7$, $p = 1.00$) despite the $+1.42$ against the baseline.

4.7 SCOPE, AND ONE CONFOUND IT CREATES

Mantri et al. (2026) report ten GLUE tasks (Wang et al., 2019) on six backbones. We report four (CoLA, STS-B, MRPC, RTE) with the full nine-arm design on BERT-base (Devlin et al., 2018), two with the same design on RoBERTa (Liu et al., 2019b), one on TinyLlama (Zhang et al., 2024), plus ModernBERT (Warner et al., 2024) and SmolLM2 (Ben Allal et al., 2025) in calibration roles only. The four tasks we chose are the four *smallest*, and not by accident: our design costs 495 runs per task, which is 22 hours for CoLA but an estimated 38 and 41 days for QQP and MNLI, with 149 GB of cached states between them.

That leaves a confound we cannot currently rule out. Every task here has at most $8{,}551$ training examples, and Stage B's failure mode is variance amplification — a quantity that shrinks as data grows. Our Stage B negative may therefore be a small-data effect rather than a property of the geometry. SST-2 (67k examples, $7.9\times$ CoLA) is the cheapest test that would settle it, and we did not run it. We do report IMDB and seven MTEB tasks (Appendix M), but only at one seed, only for the graph stage, and the MTEB rows turn out to evaluate an untrained pooler — so they carry no weight here.

5 RELATED WORK

Hyperbolic layers, and where our three stages come from. Hyperbolic space embeds trees with far less distortion than Euclidean space, which motivated Poincaré embeddings of hierarchies (Nickel & Kiela, 2017) and their Lorentz-model successor (Nickel & Kiela, 2018). Turning those embeddings into layers required a way to do linear algebra on a manifold: Ganea et al. (2018) supplied it, deriving Möbius linear maps, logistic regression and recurrent cells from gyrovector formalism, and Shimizu et al. (2021) later unified the same components with fewer parameters. Our Stage C's $W \otimes_c h = \exp_0^c(W \log_0^c h)$ is exactly that construction. Aggregation over a *neighbourhood* came next, in two flavours we reuse directly: degree-normalised tangent-space aggregation (Chami et al., 2019; Liu et al., 2019a) and attention-weighted aggregation (Zhang et al., 2019), which are our HGCN and HGAT variants. Stage B is older than either. Gulcehre et al. (2018) re-expressed soft attention in the Klein model and aggregated with the Einstein midpoint precisely because the Fréchet mean has no closed form; Chamberlain et al. (2019) scaled the same midpoint to production recommendation. In language specifically, hyperbolic word embeddings (Tifrea et al., 2018) and, recently, billion-parameter hyperbolic language models (He et al., 2025) report gains over Euclidean counterparts. What none of this literature does is vary its geometric components *independently inside one architecture*: each paper proposes a full hyperbolic model and compares it to a full Euclidean one. Our result is that on GLOT those components disagree in sign, so the aggregate comparison every one of these papers runs would have reported their average.

Measured tree-likeness as a justification. The standard empirical warrant for going hyperbolic is a small Gromov δ, computed on the data with the four-point condition, at a cost of $O(n^{3.69})$ for an exact answer (Fournier et al., 2015). Khrulkov et al. (2020) made this the explicit design step — measure δ_{rel} on a dataset's embeddings, then choose curvature accordingly — and Tifrea et al. (2018) justify hyperbolic word embeddings the same way. Our measurement in §4.5 contradicts the premise rather than the method: δ on the Euclidean metric is dominated by a handful of tokens with enormous norms, and such tokens are now a well-documented and *architectural* feature of transformers — attention sinks (Xiao et al., 2024) and massive activations (Sun et al., 2024), whose values are largely input-independent. On an angular metric that ignores them, the backbones δ is used to tell apart become indistinguishable. The same outlier and anisotropy structure (Ethayarajh, 2019; Gao et al., 2019) is what breaks the *cosine* threshold when GLOT is ported across backbones (Appendix H). The closest prior scepticism about curvature is Kochurov et al. (2020), who find Euclidean models superior on several graph tasks; they compare whole models on graph benchmarks, whereas we hold one architecture fixed and vary its parts.

Sentence pooling. Most sentence encoders reduce token states with an order-free operator and put their effort into the training objective, as with siamese fine-tuning (Reimers & Gurevych, 2019) or contrastive learning (Gao et al., 2021). A separate line builds an explicit graph over text and runs a GNN on it, classically over a corpus-level word–document graph (Yao et al., 2019). GLOT (Mantri et al., 2026) is the sentence-level, frozen-backbone instance of that idea, and the pooler we study; its components — graph attention (Veličković et al., 2018) and jumping-knowledge aggregation across layers (Xu et al., 2018) — are standard, and we also test GCN (Kipf & Welling, 2017) and GIN (Xu et al., 2019) in its place (Appendix K).

Why the protocol is half of this paper. A long line of work shows that architectural comparisons in NLP and ML are dominated by search budget and seed variance rather than by the architecture. Reimers & Gurevych (2017) showed that the seed alone moves sequence taggers by a full F1 point; Melis et al. (2017) and Lucic et al. (2018) found that carefully tuned baselines erase claimed advances in language modelling and in GANs; Musgrave et al. (2020) reached the same verdict for metric learning; and Narang et al. (2021) found that most Transformer modifications do not transfer at all. Dodge et al. (2019) give the concrete mechanism we hit twice — the maximum over T trials is a biased estimate that reorders methods as T changes — and equal-budget random search (Bergstra & Bengio, 2012) is our response to it. Our retracted Stage A result (Appendix L) is a textbook instance. What this literature does not cover, and what Appendix D adds, is that on GPU the seed itself may not identify a run, so the variance pairing is supposed to remove can be irreducible. The strongest objection to our own negative result also comes from this literature. Qiu et al. (2024) show that structured replacements for dense layers "significantly underperform" under dense-tuned learning rates and that their benefit appears only with structure-aware initialisation and step sizes. Our search moved the learning rate per arm and per task (§3), but we did not implement Riemannian optimisation or curvature-aware initialisation, so that objection stands against Stage B and we record it as such (Appendix P).

6 Conclusion

Whether hyperbolic geometry helps graph-based pooling depends entirely on *which part* receives it, and the parts disagree enough to cancel.

Stage B, the Einstein gyro-midpoint readout, is reliably harmful: negative on every task with the power to measure it, 0 or 1 of 15 seeds positive on STS-B, surviving Bonferroni on *two* encoders and repeating on a decoder, with a variance signature that suggests why. Its effect size on STS-B agrees to within 0.2 Spearman between BERT and RoBERTa. Stage C, Möbius message passing, is the one part that ever helps, by $+1.42$ MCC on CoLA and nominally positive in all four encoder settings, but it does not reach MRPC, it is significant in only one cell, and the comparison is not yet clean — measured inside a single campaign, the curvature component of that gain is $+0.07$. Stage A, the Poincaré graph, trends positive in most settings and reaches significance in none, so we leave it open. How the parts *combine* is the one thing that did not survive a second backbone.

Three lessons travel beyond this architecture. The standard reason to go hyperbolic — a low measured δ — is an artifact of norm outliers, and on an outlier-free metric the backbones it distinguishes are the same. A comparison between geometries is only as good as the budget given to its control: our own retracted result, and the confound we found in our headline number, both come from arms that were not given the same freedom. And a seed is not a run: unless deterministic kernels are requested, re-running one configuration at one seed can move a metric further than the effect being measured, which silently invalidates any table that splices cells from two campaigns. We report all three so the next study does not repeat them.

Reproducibility statement

All confirmation results use the fixed seed set $\{1, \ldots, 15\}$, shared across arms; tuning uses one held-out seed. Every run appends a row to a per-campaign CSV recording the configuration, the realised edge density, the per-epoch metrics and the wall-clock time. These logs are released unmodified, including runs that crashed or produced empty graphs, and the analysis scripts behind every table read those CSVs directly. Caches are built before any arm runs; that ordering is load-bearing, not an optimisation (§3). We do *not* claim bit-level reproducibility: Appendix D measures how far a re-run of the same configuration at the same seed can move, and the answer is far enough to matter.

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A Hyperbolic preliminaries

We work in the Poincaré ball $\mathbb{B}_c^d = \{x \in \mathbb{R}^d : c\|x\|^2 < 1\}$ of curvature $-c$, with Möbius addition $\oplus_c$ and the maps

$$\exp_0^c(v) = \tanh\big(\sqrt{c}\,\|v\|\big)\frac{v}{\sqrt{c}\,\|v\|}, \qquad \log_0^c(x) = \operatorname{artanh}\big(\sqrt{c}\,\|x\|\big)\frac{x}{\sqrt{c}\,\|x\|}, \tag{4}$$

and geodesic distance $d_c(x, y) = \frac{2}{\sqrt{c}} \operatorname{artanh}\left(\sqrt{c}\,\|-x \oplus_c y\|\right)$. With features rescaled to unit mean norm, $c = r_{\text{eff}}^2$.

B Architecture

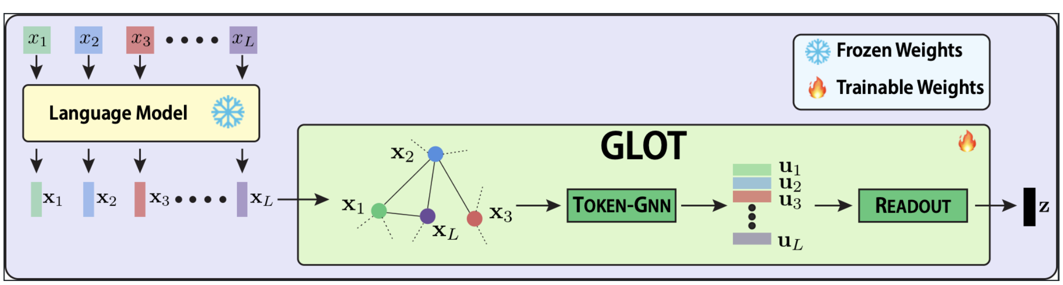

Figure 1: GLOT's pooling pipeline, reproduced from Mantri et al. (2026). Token states from a frozen backbone are thresholded into a graph (Eq. 1), refined by a token-level GNN, and aggregated into a single sentence vector $\mathbf{z}$; only the shaded block trains. Each of our three changes replaces exactly one depicted component: Stage A the graph, Stage B the readout, Stage C the token-GNN.

C Protocol

Frozen backbone, 2 epochs, Adam, weight decay searched, batch size 32; the pooler is the only trained component. Confirmation uses seeds $\{1, \ldots, 15\}$ shared across arms; tuning uses one disjoint held-out seed.

Matched budget. Trials per arm are 40 in the BERT wide campaigns and on RoBERTa, and 10 on the decoder and in the earlier graph-only campaigns. Every arm inside a given campaign gets the same number, which is the property the paired tests actually need. The decoder gets fewer because a TinyLlama run costs roughly $4\times$ a BERT run; its arms were therefore selected from a coarser search, so its entries should be read as paired deltas within that campaign and not compared against the BERT block. The cosine baseline has a smaller native space, so we widen its grid to keep selection pressure equal. Equal trials is not equal *coverage*: 40 trials cover 6.2% of a 648-point space but under 0.01% of ABC's 4.98×10^7-point space. This asymmetry works against the complex arms, and we note it as a caveat against our own negative results rather than in support of them.

Part of the grid cannot train, and almost only on one task. Pooled over every campaign with a searched learning rate (2,320 de-duplicated tuning trials), 24.3% of trials drawn at $\mathrm{lr} = 2 \times 10^{-5}$ score zero or below, against 3.5% at 2×10^{-4} and 0.1% at 1×10^{-3}. The effect is almost entirely CoLA's: on BERT/CoLA, 81 of 121 trials at that rate (66.9%) collapse to $\mathrm{MCC} = 0$ — the classifier predicts the majority class — as do 14 of 120 at 2×10^{-4}, so 26% of that task's nominal 40-trial budget buys nothing. RoBERTa/CoLA behaves the same way and slightly worse (93 of 130, 71.5%), while MRPC sits at 2.3% and STS-B and RTE at exactly 0% on both backbones. Every arm draws from the same grid, so no comparison is biased by this; but the effective search on CoLA is roughly a quarter smaller than the headline number, and that is the task where our one positive result lives.

Testing. We report the exact two-sided sign test beside the paired t-test because evaluation-set quantisation makes the parametric standard error look too small (MRPC accuracy moves in steps of 0.245 on 408 examples). Both tests must agree. Bonferroni is applied over the arms tested against a common baseline *within* a campaign: 8 arms on the BERT campaigns, giving $\alpha = 0.0063$; 10 on RoBERTa, where the seven hyperbolic arms share a baseline with the three calibration arms, giving $\alpha = 0.005$; and 6 on the decoder, giving $\alpha = 0.0083$. The count is a property of the campaign, so adding arms to a campaign tightens the threshold for arms already in it — which it did once here (Appendix H). Minimum detectable effects at $n = 15$, from the median paired standard error actually observed, are 0.225 (BERT STS-B), 1.153 (BERT CoLA), 0.317 (MRPC), 2.064 (RTE), 1.000 (RoBERTa CoLA), 0.243 (RoBERTa STS-B) and 0.503 (TinyLlama). RTE's figure exceeds every effect in this paper, so that task can only ever return a null; we report it, and draw nothing from it.

D A seed does not pin a run

`set_seed` in the released code seeds `random`, `numpy`, `torch` and `torch.cuda`. It does not call `torch.use_deterministic_algorithms`, does not set `cudnn.deterministic`, and does not set `CUBLAS_WORKSPACE_CONFIG`. Scatter-based neighbourhood aggregation uses atomic adds on CUDA, whose summation order is not fixed, so two runs of the same configuration at the same seed diverge from the first backward pass and the divergence is amplified over two epochs of training.

We measured it by accident. The extended Stage C factorial re-runs the baseline cell itself rather than borrowing it, and its configuration string is character-for-character identical to the one recorded for the `baseline` arm of the main CoLA campaign. Over the 15 shared seeds:

Statistic of (factorial re-run − original), CoLA MCC	value
seeds where the two agree exactly	0 of 15
mean difference	$+0.334$
standard deviation of the difference	1.472
largest single-seed difference	3.61
for comparison: the effect this campaign reports for Stage C	$+1.416$

Two consequences, one benign and one not.

Benign: within-campaign paired tests are still valid. Both arms of every paired test in this paper were run in the same campaign, once each. Re-run non-determinism is part of the within-pair scatter that the paired t and sign tests already estimate from the data; it inflates the standard error rather than biasing the difference. Nothing in Tables 4–5 needs to change.

Not benign: pairing recovers less than we claimed, unevenly. On BERT STS-B the paired difference for arm B has $\mathrm{sd} = 0.243$ against an independent-samples estimate of 0.489 — pairing halves the variance, as intended. On BERT CoLA the paired difference for arm C has $\mathrm{sd} = 2.08$ against an independent-samples estimate of 1.52: pairing is *worse* than not pairing, which means the seed explains essentially none of CoLA's variance and the run-to-run term dominates. Statements of the form "every comparison is paired, so the dominant variance component cancels" are true on STS-B and false on CoLA.

And: cells may not be spliced across campaigns. Our own 2×2 in Table 6 originally took two cells from the wide campaign and ran only the two off-diagonal ones. Substituting the re-run of the shared cell changes the geometry component of Stage C's gain from $+0.40$ to $+0.07$ (Table 7). The in-flight 65-seed job runs all four cells through one script, one code path and one cache state for this reason.

E SELECTED HYPER-PARAMETERS

Each arm tunes its own configuration; these are the winners then run on all 15 confirmation seeds. Two things are worth reading off Table 1. The learning rate does not converge to a single value across tasks, so no arm was pinned to a rate chosen for a flat pooler. And on CoLA, Stage C selected a much denser graph ($q = 0.38$) and a smaller network than the baseline ($q = 0.10$, twice the depth, twice the width) — the confound discussed in §4.3. On MRPC the same search went the other way, giving C a $15\times$ *sparser* graph than the baseline; C is negative there. C wins where it picked dense and loses where it picked sparse, which is why the matched comparison is necessary.

Table 1: Confirmed configurations, BERT wide campaigns. "q" is `tau_quantile` for cosine arms and `rho_quantile` for Poincaré arms; both mean "keep the closest q fraction of token pairs". "width" is the GNN hidden dimension, "proj" the projection dimension.

Task	Arm	lr	layers	width	proj	q
CoLA	baseline	1×10^{-3}	4	256	128	0.10
	A	1×10^{-3}	4	256	256	0.05
	B	2×10^{-4}	2	256	512	0.38
	C	1×10^{-3}	2	128	256	**0.38**
	AC	1×10^{-3}	2	128	256	0.05
STS-B	baseline	2×10^{-4}	4	64	256	0.20
	A	2×10^{-4}	4	64	256	0.10
	B	2×10^{-4}	4	64	128	0.38
	C	2×10^{-4}	2	128	128	0.20
	AC	2×10^{-4}	4	128	512	0.20
MRPC	baseline	2×10^{-4}	2	128	128	0.38
	B	1×10^{-3}	4	128	256	0.10
	C	2×10^{-4}	2	128	256	**0.025**

Other selected settings vary similarly. Jumping-knowledge mode is uniform across all nine arms on CoLA (`max`) and on STS-B (`cat`), and near-uniform on MRPC and RTE, where exactly one arm each departs from the majority — so it tracks the task far more than the arm. Curvature, where it applies, spans the whole searched range $\{0.25, 1.0, 4.0\}$ with no consistent preference.

F REPRODUCTION

Table 2: One method, four estimators, on CoLA. Each row is the *same* pooler; only the way the score is computed changes. A dash means the knob does not apply. The published value sits between our mean and our maximum, and the row that matches how it was computed exceeds it.

Seeds	Aggregated	Selected over	CoLA MCC
Published (Mantri et al., 2026)			
1 (seed 42)	—	best τ of 3	47.49
Ours, same method			
15	mean	—	45.37
15	max	—	48.17
1 (seed 42)	—	best of 40 configs	**50.16**

Table 3: Matched-estimator comparison: best score at GLOT's own seed (42) over GLOT's own grid, 78 configurations per task, so both columns are computed the same way.

Task	ours	published	Δ
CoLA (MCC)	**50.52**	47.49	$+3.03$
RTE (acc.)	**59.57**	59.21	$+0.36$
STS-B (Spearman)	82.99	83.86	-0.87

The published τ-sensitivity curve (their Table 8) cannot be read as a τ effect: it spans 7.87 MCC on CoLA, while we measure a spread of 5.46 across seeds at *fixed* τ.

G MAIN RESULTS

Table 4 gives the absolute scores behind every delta in this paper. It follows the layout of Table 1 of Mantri et al. (2026) — backbone blocks, method rows, task columns, scores $\times 100$, best per column in bold, metric row under the header — but the rows are our arms rather than competing poolers, because our question is which *component* of GLOT does what, not whether GLOT beats mean-pooling. The four bag-of-tokens rows of the original (`[CLS]`/`[EOS]`, Mean, Max, AdaPool) are not filled: we never ran them on GLUE. Those poolers appear only in the noise diagnostic (Appendix J), where they lose by 14–35 points.

Two things are visible here that the delta tables hide. The whole spread on MRPC is 0.37 F1 across the nine arms run on it, and on RTE the seed standard deviation (1.5–3.4) is larger than every gap in the column — which is why we treat both as null. And within the BERT block the bolded best cell names four *different* arms across four columns, which is what a table of noise looks like.

Paired statistics, BERT STS-B: B $\bar{d} = -1.065$, $t = -17.00$, sign $p = 6 \times 10^{-5}$; BC -1.242, $t = -8.30$, $p = 6 \times 10^{-5}$; ABC -0.837, $t = -6.36$, $p = 0.00098$; AB -0.747, $t = -5.66$, $p = 6 \times 10^{-5}$. All four survive Bonferroni. BERT CoLA: C $\bar{d} = +1.416$, $t = 4.11$, sign $p = 0.0074$ — significant on both tests but above $\alpha = 0.0063$. RoBERTa STS-B: B $\bar{d} = -1.264$, $t = -15.76$, $p = 6 \times 10^{-5}$; AB -0.795, $t = -5.77$, $p = 0.00098$; published τ $+0.321$, $t = 3.13$, $p = 0.035$ — significant on both tests uncorrected, not after Bonferroni. RoBERTa CoLA: BC -2.161, $t = -6.20$, $p = 6 \times 10^{-5}$; ABC -3.314, $t = -5.44$, $p = 0.00098$; quantile τ -1.408, $t = -3.44$, $p = 0.0074$. TinyLlama STS-B: BC -1.165, $t = -4.97$, $p = 0.00098$; AB -3.556, $t = -6.95$, $p = 6 \times 10^{-5}$; ABC -4.506, $t = -4.85$, $p = 6 \times 10^{-5}$. Every arm in these tables has exactly 15 seeds; shard files and merged campaign files contain the same rows, so the analysis de-duplicates by run key before testing.

Table 4: Absolute confirmed scores, mean ± sd over the 15 shared seeds, ×100. Metrics follow Mantri et al. (2026): MCC for CoLA, Spearman for STS-B, F1 for MRPC, accuracy for RTE. Best per column within a backbone block in bold. `no graph` deletes the token graph entirely and is an ablation of GLOT itself, not one of our stages; `published` τ and `quantile` τ are RoBERTa threshold-calibration arms (Appendix H). Every number is a confirmation mean, never a tuning maximum. A dash is a cell we do not intend to run; ?? is a cell we do.

Backbone	Method	CoLA MCC ↑	STS-B Spear. ↑	MRPC F1 ↑	RTE Acc. ↑
BERT-base	GLOT (baseline)	45.37 ±1.30	82.53 ±0.33	82.14 ±0.44	53.79 ±3.36
	A	46.07 ±0.99	82.56 ±0.31	**82.18** ±0.29	53.82 ±1.58
	B	44.11 ±1.48	81.47 ±0.36	81.81 ±0.39	54.78 ±1.58
	C	**46.78** ±0.79	82.73 ±0.31	81.83 ±0.39	54.35 ±2.07
	AB	44.13 ±1.61	81.79 ±0.46	81.86 ±0.41	54.59 ±2.93
	AC	45.82 ±1.94	**82.77** ±0.20	81.99 ±0.35	55.26 ±1.80
	BC	45.59 ±1.19	81.29 ±0.32	81.83 ±0.43	**55.64** ±1.88
	ABC	44.84 ±1.87	81.70 ±0.34	81.88 ±0.44	54.85 ±1.52
	`no graph`	46.20 ±2.54	82.57 ±0.34	82.17 ±0.31	53.86 ±2.74
RoBERTa-base	GLOT (baseline)	54.60 ±1.52	83.89 ±0.30	—	—
	A	54.37 ±0.89	83.80 ±0.30	—	—
	B	54.23 ±1.64	82.62 ±0.41	—	—
	C	**55.25** ±1.39	84.18 ±0.40	—	—
	AB	53.89 ±1.71	83.09 ±0.43	—	—
	AC	54.60 ±1.31	83.95 ±0.28	—	—
	BC	52.44 ±1.53	83.63 ±0.41	—	—
	ABC	51.29 ±1.83	83.75 ±0.30	—	—
	`no graph`	54.32 ±2.06	83.84 ±0.32	—	—
	published τ	54.46 ±1.18	**84.21** ±0.29	—	—
	quantile τ	53.19 ±1.59	83.73 ±0.29	—	—
TinyLlama-1.1B	GLOT (baseline)	??	79.95 ±0.49	—	—
	A	??	79.99 ±0.51	—	—
	B	??	??	—	—
	C	??	??	—	—
	AB	??	76.39 ±1.87	—	—
	AC	??	**80.15** ±0.54	—	—
	BC	??	78.78 ±0.95	—	—
	ABC	??	75.44 ±3.58	—	—
	`no graph`	??	79.95 ±0.48	—	—

G.1 DECOMPOSING STAGE C

Stage C's selected configuration differed from the baseline's in two ways at once: a 3.6× denser graph ($q = 0.38$ against 0.10) and a smaller network (2 layers of width 128 against 4 of 256). We therefore ran the 2×2 that crosses *message-passing geometry* with *graph-and-architecture configuration*, on the same 15 seeds. The hyperbolic cells use C's own tuned curvature settings ($c = 4.0$, HGAT, input clip 0.7), which is the choice most favourable to C.

Each component differences two independently trained cells, so it inherits both their variances; that is why the parts are noisier than the whole. At the observed effect and spread, detecting the interaction at 80% power would take about 65 seeds. This is the cleanest illustration we have of two general hazards at once: a significant end-to-end improvement can be genuinely irreducible at the sample size that established it, and the apparent size of its components can depend on which run of a shared cell you subtract.

[TODO: The 65-seed job runs all four cells. When it lands, delete the "spliced" column and Table 6's italic row, and rebuild both tables from that file alone.]

Table 5: Per-stage verdict: paired mean differences against each backbone's own GLOT baseline on shared seeds, with the seed split positive/negative. † significant on both the paired t-test and the exact sign test; ‡ additionally surviving that campaign's Bonferroni correction ($\alpha = 0.0063$ on BERT, 0.005 on RoBERTa, 0.0083 on the decoder). *No RTE entry is significant and none could be*: that task has 277 evaluation examples and a minimum detectable effect of 2.06 accuracy points, larger than any effect in this paper.

Backbone	Method	CoLA	STS-B	MRPC	RTE
BERT-base	A	$+0.70$ (11/4)	$+0.03$ (9/5)	$+0.04$ (8/7)	$+0.02$ (6/9)
	B	-1.26 (5/10)	$-1.07^{\ddagger}$ (0/15)	-0.33 (5/10)	$+0.99$ (8/7)
	C	$+\mathbf{1.42}^{\dagger}$ (13/2)	$+0.20$ (12/3)	-0.32 (4/11)	$+0.55$ (8/7)
	AB	-1.24 (4/11)	$-0.75^{\ddagger}$ (1/14)	-0.28 (5/10)	$+0.79$ (7/6)
	AC	$+0.45$ (10/5)	$+\mathbf{0.24}$ (11/4)	-0.15 (6/8)	$+1.47$ (11/4)
	BC	$+0.22$ (7/8)	$-1.24^{\ddagger}$ (0/15)	-0.31 (5/10)	$+\mathbf{1.85}$ (11/4)
	ABC	-0.53 (6/9)	$-0.84^{\ddagger}$ (1/14)	-0.26 (4/11)	$+1.06$ (8/6)
	`no graph`	$+0.83$ (11/4)	$+0.03$ (7/8)	$+0.03$ (9/6)	$+0.07$ (7/6)
RoBERTa-base	A	-0.23 (5/10)	-0.09 (5/10)	—	—
	B	-0.37 (5/10)	$-1.26^{\ddagger}$ (0/15)	—	—
	C	$+\mathbf{0.64}$ (8/7)	$+0.29$ (10/5)	—	—
	AB	-0.71 (5/10)	$-0.79^{\ddagger}$ (1/14)	—	—
	AC	$+0.00$ (6/9)	$+0.06$ (7/8)	—	—
	BC	$-2.16^{\ddagger}$ (0/15)	-0.26 (4/11)	—	—
	ABC	$-3.31^{\ddagger}$ (1/14)	-0.13 (8/7)	—	—
	`no graph`	-0.29 (5/10)	-0.05 (7/8)	—	—
	published τ	-0.14 (8/7)	$+\mathbf{0.32}^{\dagger}$ (12/3)	—	—
	quantile τ	$-1.41^{\dagger}$ (2/13)	-0.15 (5/10)	—	—
TinyLlama-1.1B	A	??	$+0.04$ (9/6)	—	—
	B	??	??	—	—
	C	??	??	—	—
	AB	??	$-3.56^{\ddagger}$ (0/15)	—	—
	AC	??	$+\mathbf{0.21}$ (10/5)	—	—
	BC	??	$-1.17^{\ddagger}$ (1/14)	—	—
	ABC	??	$-4.51^{\ddagger}$ (0/15)	—	—
	`no graph`	??	$+0.00$ (8/7)	—	—

On BERT, every arm containing B is negative on CoLA, STS-B and MRPC. All four are positive on RTE, none significantly.
Every arm containing C but not B is positive on CoLA and STS-B, on both encoders.
The three STS-B columns agree in sign on all five arms run on all three backbones.

Table 6: Density × curvature factorial, CoLA MCC, mean ± sd over the 15 shared seeds. Realised edge density was verified per cell: 0.0999 and 0.0998 in the base-config row, 0.3758 in the C-config row, so the rows really are density-matched. Three of the four cells were run by the factorial script itself; the bottom-right cell is still the `C` arm of the main campaign, which Appendix D shows is not a safe substitute for a re-run. The italic row is the value the earlier revision of this paper used for the top-left cell.

	Euclidean MP	hyperbolic MP
base config (q=0.10, K=4, h=256)	45.70 ±1.87	45.77 ±0.90
same cell, spliced from the main campaign	*45.37 ±1.30*	—
C config (q=0.38, K=2, h=128)	45.36 ±2.05	**46.78** ±0.79

H Porting to other backbones

The failure is layer-dependent: ModernBERT reaches 0.599 at the final layer but 0.996 at layer 12. Mantri et al. (2026) do not state which layer they pool, so "which backbones are affected" cannot

Table 7: Decomposition of Stage C's headline CoLA gain, paired over 15 seeds. Within each column the three components sum exactly to the total. The two columns differ only in which draw of the base-config Euclidean cell is subtracted. No component in either column is significant on either test.

	within one campaign				spliced (earlier revision)			
Component	Δ	sd	t	p/n	Δ	sd	t	p/n
geometry alone (at base config)	$+0.066$	2.40	0.11	7/8	$+0.400$	1.75	0.88	8/7
configuration alone (Euclidean MP)	-0.346	2.67	-0.50	8/7	-0.012	2.55	-0.02	10/5
interaction	$+1.362$	3.17	1.66	11/4	$+1.028$	2.95	1.35	10/5
total (C $-$ base cell)	$\mathbf{+1.082}$	1.92	2.18	12/3	$\mathbf{+1.416}$	1.33	$\mathbf{4.11}$	**13/2**

Table 8: Token-cosine percentiles and the edge density the published grid $\tau \in \{0.1, 0.3, 0.6\}$ actually yields. Measured on 192 CoLA validation sentences at the final layer, `max_length` 128, off-diagonal pairs only, self-loops excluded. This is a property of the backbone and τ alone. Density near 1 means the GNN sees a nearly complete graph.

Backbone	cos p10	cos p50	cos p90	τ=0.1	τ=0.3	τ=0.6
TinyLlama-1.1B	0.016	0.343	0.588	0.762	0.533	**0.073**
ModernBERT-base	0.076	0.648	0.768	0.866	0.759	0.599
BERT-base	0.098	0.328	0.595	0.876	0.505	**0.103**
SmolLM2-360M	0.322	0.774	0.911	0.956	0.908	0.741
RoBERTa-base	**0.693**	0.821	0.897	1.000	1.000	**0.991**

be settled from the published description; we treat RoBERTa as the one clear case. This table is also why we chose TinyLlama over SmolLM2 as the decoder: its density at the published $\tau = 0.6$ (0.073) is close to BERT's (0.103), whereas SmolLM2 sits at 0.741.

Table 9: RoBERTa-base calibration arms, $n = 15$ paired seeds, tuned per arm and per task on RoBERTa itself (40 trials each). `published_tau` pins the published grid and gives the near-complete graph above; `quantile_tau` is the density-matching repair. The repair is the worst of these four on both tasks — despite having *won* the tuning stage (56.77, best of five). The hyperbolic arms tuned on the same backbone are in Table 5.

	CoLA (MCC)			STS-B (Spearman)		
Arm	mean	Δ	p/n	mean	Δ	p/n
baseline	**54.60**	*ref*	—	83.89	*ref*	—
`published_tau`	54.46	-0.14	8/7	**84.21**	$\mathbf{+0.32}$	12/3
A	54.37	-0.23	5/10	83.80	-0.09	5/10
`quantile_tau`	53.19	$\mathbf{-1.41}$	2/13	83.73	-0.15	5/10

Forcing the graph sparse costs 1.41 MCC ($t = -3.44$, 2/13, sign $p = 0.0074$). An earlier revision reported this as surviving Bonferroni at $\alpha = 0.0125$, which was correct when only the four calibration arms were tested against this baseline. The RoBERTa campaign now puts ten arms against the same baseline, so the correction tightens to $\alpha = 0.005$ and the repair's deficit no longer survives it. The effect is unchanged; only the number of comparisons it must be judged against. Against `published_tau` directly the repair loses on both tasks (-1.27 CoLA, -0.48 STS-B, the latter $2/13$, $p = 0.0074$).

Density and scale are separate. We had credited a $21.71 \rightarrow 27.15$ MCC recovery on ModernBERT/CoLA to density-matching plus median-norm rescaling, measured at one seed with both factors moved together. The 2×3 factorial over 5 seeds (Table 10) separates them. The choice of rescaling statistic matters far more than whether you rescale, and it is the only effect here significant on both tests: on ModernBERT layer 12, whose mean/median norm ratio is 7.66, mean-based (`rms`) rescaling costs 11.03 MCC ($t = -5.47$, $0/5$) — the mean is exactly the statistic outliers corrupt. Median rescaling is directionally positive (+5.96 and +5.32 on layer 12, +1.34 at the final layer with $5/0$) but reaches significance nowhere at $n = 5$. Density-matching has no consistent sign (+4.58, −1.91, −0.25, +0.47). BERT is inert throughout, every contrast at most 0.27 MCC — our earlier single-seed reading of that control ($45.54 \rightarrow 43.41$) was noise.

Table 10: Density × scale factorial, CoLA MCC, mean ± sd over 5 seeds. Columns vary the threshold, rows vary token-norm rescaling. At $n = 5$ the exact sign test cannot fall below $p = 0.0625$, so nothing here can meet our both-tests bar; we report it as a lead.

Backbone (layer)	Rescaling	$\tau = 0.6$ (pub.)	$q = 0.05$
BERT-base (final) *control*	none	45.25 ±1.09	45.00 ±1.69
	`rms`	45.01 ±1.39	45.03 ±1.32
	`median`	44.98 ±1.12	45.08 ±0.93
ModernBERT-base (L12)	none	18.64 ±11.36	23.22 ±6.90
	`rms`	7.61 ±8.62	19.18 ±3.33
	`median`	24.60 ±7.96	**28.54** ±2.69
ModernBERT-base (final)	none	37.53 ±1.21	35.62 ±6.70
	`rms`	38.64 ±1.66	37.61 ±1.92
	`median`	**38.87** ±1.32	37.57 ±2.00
RoBERTa-base (final)	none	52.39 ±2.33	**52.87** ±0.96
	`rms`	52.55 ±1.55	52.63 ±1.42
	`median`	52.54 ±1.41	52.72 ±1.40

I Decoder backbone

TinyLlama-1.1B on STS-B, 10 tuning trials per arm and $n = 15$ shared confirmation seeds. The trial budget is lower than the BERT campaigns' 40 because a run here costs about four times as much; it is identical across the arms in this campaign, which is what the paired tests require, but it does mean these arms were selected from a coarser search than their BERT counterparts. Backbone chosen over SmolLM2 on measured graph density (Table 8). Minimum detectable effect 0.503. The absolute scores are in Table 4 and the deltas in Table 5; the ordering matches BERT, and the standard deviation grows in step with the number of hyperbolic parts ($0.49 \rightarrow 0.95 \rightarrow 1.87 \rightarrow 3.58$).

[TODO: Two gaps. B and C *alone* were dropped from this sweep for cost, so the paper currently claims a decoder replication of Stage B without ever having run B by itself on a decoder. And the decoder CoLA campaign has finished its tuning stage but has *no* confirmation rows, so the whole CoLA column of the TinyLlama block is empty — it will be the first test of Stage C on any backbone other than BERT and RoBERTa, which is the single largest gap in the paper's one positive result.]

J Noise diagnostic

We reproduce the signal-in-noise test of Mantri et al. (2026) (their Table 7): a short signal phrase carrying a logical dependency, injected into random distractor words, with the distractor ratio swept from 20% to 90%. BERT, mean ± sd over 5 seeds; the original reports one seed. We follow their layout: methods as rows, distractor ratios as columns, best per column in bold.

Three readings, in decreasing order of confidence.

GLOT's advantage over bag-of-tokens pooling is real and holds in every seed. Mean, `[CLS]` and Max lose 14–35 points at every ratio, $0/5$ seeds positive. This is the claim the original rests on and we reproduce it.

Table 11: Diagnostic stress test, BERT-base, accuracy $\times 100$, mean $\pm$ sd over 5 seeds. The upper block is our hyperbolic arms, the middle block the controls, the lower block the poolers of the original. Stage A appears twice: the Poincaré graph can be built by thresholding distance at a quantile or by k-NN, and the stress harness supports both.

Method	20%	50%	80%	90%
A (k-NN)	97.0 $\pm$1.3	91.9 $\pm$4.1	91.0 $\pm$2.7	84.9 $\pm$9.1
A (quantile)	96.5 $\pm$1.0	91.5 $\pm$6.5	89.8 $\pm$5.7	83.1 $\pm$8.2
B	95.4 $\pm$1.4	88.7 $\pm$2.8	83.8 $\pm$13.8	75.5 $\pm$14.9
C	95.0 $\pm$3.3	87.6 $\pm$11.2	90.8 $\pm$5.9	88.4 $\pm$8.8
AC	95.0 $\pm$3.4	88.2 $\pm$11.2	90.8 $\pm$5.8	**88.5** $\pm$8.9
ABC	88.7 $\pm$12.3	82.5 $\pm$15.4	86.0 $\pm$10.5	84.5 $\pm$8.9
GLOT (baseline)	96.0 $\pm$0.5	90.2 $\pm$5.0	88.1 $\pm$4.7	79.8 $\pm$10.8
`no graph`	**98.0** $\pm$1.3	**94.8** $\pm$5.7	**92.5** $\pm$5.0	84.4 $\pm$10.1
GLOT K=0	95.7 $\pm$1.6	91.0 $\pm$3.6	89.3 $\pm$5.9	88.1 $\pm$6.5
AdaPool	95.1 $\pm$1.2	90.8 $\pm$2.7	86.5 $\pm$6.4	83.6 $\pm$8.4
Mean	73.5 $\pm$3.2	63.2 $\pm$5.2	64.0 $\pm$1.6	57.4 $\pm$3.2
`[CLS]`	70.0 $\pm$1.7	61.4 $\pm$2.9	62.1 $\pm$3.2	65.4 $\pm$2.4
Max	63.0 $\pm$3.4	55.2 $\pm$2.8	53.1 $\pm$1.8	51.8 $\pm$2.6
Mantri et al. (2026) Table 7, BERT rows (single seed):				
GLOT	97.2	97.0	97.8	98.8
AdaPool	91.4	78.8	65.6	61.6

But the graph is not what produces it. `no graph` — the same trained pooler with every edge deleted — is the best arm at three of the four ratios, and GLOT with K=0 GNN layers beats GLOT at 80% and 90%. Whatever survives the noise here is the trained projection and readout, not message passing. Stage A is ahead of the baseline at all four ratios but behind `no graph` at three; Stage B is last of the hyperbolic arms, consistent with §4.2. An earlier revision of ours reported "Stage A wins the stress test"; with the control in the table, the honest statement is that removing the graph wins.

We do not reproduce the published trained-pooler numbers. At 90% distractors the original reports 98.8 where we measure 79.8 $\pm$ 10.8 for the same method, and their curve is flat or rising under noise where ours falls. The divergence is not uniform: the static poolers agree closely (-2.2 to $+5.6$ across twelve cells), while both *trained* poolers diverge in opposite directions, so the 37-point margin their Table 7 reports for GLOT over AdaPool at 90% becomes -3.9 in our hands. Their Appendix B.4 does not give the vocabulary source, the signal templates, or the pooler budget, so we cannot localise the cause. We flag this as unresolved, and we never use the published values as a baseline.

K GNN CHOICE AND TRAINING RECIPE

GNN backbone. Over 5 seeds at the published recipe, CoLA gives GAT 45.38 $\pm$ 1.19, GCN 45.14 $\pm$ 1.78, GIN 44.86 $\pm$ 1.33; STS-B gives 82.69 $\pm$ 0.34, 82.46 $\pm$ 0.27, 82.47 $\pm$ 0.32. Spread 0.52 MCC and 0.23 Spearman, inside seed noise, and GAT is the only one of the three that consumes edge attributes at all.

The STS-B residual is not a recipe mismatch. The released README trains differently from the paper's Appendix B.2 (3 epochs, `scorer_hidden` 256, $\tau = 0.8$ against 2, 128, 0.6). We ran the $2 \times 2 \times 2$ decomposition at seed 42. Epoch count is the only factor that moves anything (82.36 $\rightarrow$ 82.66); τ and `scorer_hidden` are inert to within 0.04. Over 5 seeds the two recipes are indistinguishable (82.69 $\pm$ 0.34 against 82.69 $\pm$ 0.23). The best cell, 82.66, is still 1.20 below the published 83.86, so the residual stands.

Layer choice overfits. Layer 8 beat layer 12 by $+4.62$ MCC on CoLA but only $+0.25$ on STS-B, and lost on RTE (-0.36) and MRPC (-0.46).

L A RETRACTED RESULT OF OUR OWN

Under the graph-only search, Stage A on STS-B gave $\bar{d} = +0.223$, $t(14) = 6.20$, sign $p = 6\times10^{-5}$, $15/15$ seeds positive, and replicated on 12 held-out seeds. It does not survive the wide search: $\bar{d} = +0.028$, $t = 0.69$, $9/5$. The effect was real but its cause was not geometry — the baseline had been pinned at $K = 2$, $h = 128$ while nothing else was, and it recovers the whole gap once it can choose $K = 4$. This is the clearest illustration of the paper's own point: a pooling comparison is only as good as the budget given to its control.

M LONG-TEXT AND RETRIEVAL BENCHMARKS

Mantri et al. (2026) also report IMDB (long-text classification, their Table 2) and MTEB (Muennighoff et al., 2022). We ran both, early in the project, at a single seed and on BERT only. We report them here for coverage and draw no statistical conclusion: with $n = 1$ there is no test to run.

Two caveats that make this section evidence of nothing. First, the MTEB rows evaluate an *untrained* pooler. Mantri et al. (2026) obtain their MTEB numbers by contrastively training the pooler on MS MARCO triplets and then evaluating zero-shot. *We never ran that stage.* Our driver never sets `checkpoint_path`, and the evaluation entry point loads pooler weights only when that argument is set, so every MTEB row below was produced by a pooler at its random initialisation on top of the frozen backbone. The wall-clock times confirm it: 7–171 seconds per task, which is inference and nothing else. The three MTEB columns therefore differ only by random initialisation — they agree to the third decimal on five of seven tasks — and carry no information about Stage A whatsoever. Second, the seven tasks below are ones we selected from the original's *appendix* Table 12, not the seven of its Table 3 (EmotionClassification, SciFact, RedditClustering, AskUbuntu, STS12, TwitterSemEval, SummEval); only SciFact and STS12 are common to both. We leave the table in place because an earlier draft described these rows as trained and as matching the original's main table, and silently deleting it would hide two corrections. The IMDB rows are unaffected: that task trains the pooler normally.

One of these rows is not what its name says. These runs predate our switch to quantile thresholding. The ρ-defect arm used an absolute radius $\rho = 1.0$ on unnormalised BERT features, and as Appendix N explains, real Poincaré distances there lie in $[8.85, 11.76]$ — so that arm trained on an empty graph and is *not* a Stage A result. We keep the column only so it cannot be mistaken for one. The k-NN column is a genuine Stage A variant.

Across seven MTEB tasks the largest gap to the baseline is 2.34, on a task where all three columns sit within a few points of each other — which is what a random pooler evaluated three times should look like. Nothing here is separable at one seed, and neither Stage B nor Stage C was ever run on these benchmarks. Doing it properly is expensive on the IMDB side: our runs took about 3.8 hours each at 512 tokens, so eight arms $\times$ 15 seeds would be roughly 20 days. The MTEB side is cheap to *evaluate* (all seven tasks take about eight minutes) but requires building the MS MARCO contrastive stage first.

[TODO: The corrected MTEB pipeline exists and has been validated end to end but has not been launched. When it lands, replace this table rather than amending it, use the original's Table 3 task list so the comparison is like-for-like, and note that $n = 5$ cannot reach our both-tests bar — the exact sign test bottoms out at $p = 0.0625$ — so the result will be a lead, of the same status as Table 10.]

N NOTES FOR REPRODUCERS

Five details of the released code cost us time. The first three do not affect the conclusions of the original paper; the last two are asymmetries between our own arms.

Table 12: IMDB accuracy and MTEB main scores ($\times 100$), BERT-base, seed 42, one run per cell. The middle column is the ρ-defect run described above; it trained on an empty graph and is not a Stage A measurement. **The MTEB block evaluates an *untrained* pooler** — we did not run the MS MARCO contrastive stage — so those three columns differ only by random initialisation and measure nothing about the method.

	GLOT (cosine)	ρ-defect† (empty graph)	Stage A (k-NN)
IMDB (acc.)	92.26	**92.50**	92.46
Banking77Classification	46.91	46.90	**46.91**
STS12	30.02	**30.04**	27.68
STS13	59.57	59.60	**59.83**
SciFact	13.93	**14.10**	13.39
ArguAna	**28.25**	28.11	27.73
TwentyNewsgroupsClustering	24.49	24.13	**25.10**
SprintDuplicateQuestions	**34.67**	34.57	34.40
MTEB mean	**33.98**	33.92	33.58

- `jk_mode` is advertised as {`cat`, `max`, `mean`, `none`}, the CLI accepts {`cat`, `lstm`, `max`}, and the pooler implements {`cat`, `max`, `mean`}. `lstm` parses and then raises at runtime; `mean` is implemented but unreachable from the CLI. The usable space is {`cat`, `max`}.
- Thresholding Poincaré *distance* at a fixed radius gives an empty graph on real features (see §2). The signature is a score that is bit-identical across a whole threshold grid. We use a quantile threshold everywhere for this reason.
- The logged edge-density counts self-loops in the numerator but excludes the diagonal from the denominator, so an *empty* graph logs $1/(n-1) \approx 0.10$ at CoLA's typical length rather than 0. This is a reporting defect only — no training path reads the statistic — but it is a trap for exactly the audit this paper performs. It was corrected partway through the project, so the CoLA logs carry the offset and the later MRPC logs do not; densities are not comparable across campaigns without checking. Any density we quote is re-measured with the standalone instrument of Table 8, never taken from training telemetry.
- **Our Stage C layers add self-loops twice.** The graph builder emits self-loops, and `HyperbolicGCNConv`/`HyperbolicGATConv` then call `add_self_loops` without removing existing ones first, so every node gets two. PyG's Euclidean `GATConv`, which the baseline uses, removes before adding. Measured on the real BERT/CoLA cache (105 nodes, 587 edges) the hyperbolic layers see 692 edges where the Euclidean one sees 587. Forward-pass effect at Stage C's tuned configuration: relative output change 2.38×10^{-2} at q=0.38 and 1.92×10^{-2} at q=0.10, against 4.13×10^{-1} for a dense-to-sparse graph change and exactly 0 for the same call twice. It is small, but it is real, and because it differs between the two hyperbolic cells it does not cancel in the "geometry alone" contrast of Table 7. **[TODO: Re-run the two hyperbolic cells with the duplicate removed (≈1.5 GPU-hours) before, not after, the 65-seed factorial; otherwise that job measures the buggy layer at higher precision.]**
- `HyperbolicGATConv` accepts `edge_weight` and never uses it, so `--edge_weight_mode` is a no-op whenever the hyperbolic GAT is selected. The Euclidean `GATConv(edge_dim=1)` does consume it. No arm in this paper varies that flag, so no reported comparison is affected, but the released grid contains a knob that does nothing for half the arms.

O AUDIT TRAIL FOR THIS REVISION

Every table was recomputed from the campaign CSVs. Fourteen values changed. They are listed here rather than corrected silently, because several of them are the kind of error this paper criticises in others.

- RTE's minimum detectable effect was quoted as 2.7 accuracy points; from the wide campaign's own paired scatter it is 2.064. It appeared in five places. The conclusion drawn from it — RTE cannot resolve any effect in this paper — is unchanged.
- The pooled share of tuning trials that fail to train was quoted as $19.2\%/2.2\%/0\%$ across the three learning rates; recomputed over all 2,320 de-duplicated tuning rows it is $24.3\%/3.5\%/0.1\%$. The per-task figures (81 of 121 on BERT/CoLA, 66.9%) were correct. The RoBERTa/CoLA figure was quoted as 44.1% and is 71.5%.
- Two standard deviations in Table 6 were wrong ($1.42 \to 0.90$, $2.19 \to 2.05$).
- Table 10's BERT and ModernBERT blocks used population standard deviations while its RoBERTa block used sample standard deviations from a different file. All are now sample standard deviations.
- The decoder's baseline seed standard deviation was quoted as 0.48 and is 0.494.
- The noise-diagnostic table labelled one row "A" when two Stage A variants exist and the row shown was the k-NN one. Both are now shown.
- The same table omitted `no graph`, GLOT $K{=}0$ and ABC. `no graph` is the best arm at three of four ratios, which reverses the paragraph that accompanied the table.
- "Stage A is ahead in the noise diagnostic at three of four ratios" was wrong in our own favour: it is ahead of the *baseline* at four of four, and behind `no graph` at three of four.
- "On MRPC every hyperbolic arm is slightly negative" was wrong: Stage A is $+0.04$. The claim now names the arms containing B or C.
- The paper stated that the `no graph` control had never been run on BERT and listed it as an unscheduled experiment. It had been run, on all four BERT tasks and on the decoder, and it is mildly positive on BERT — which strengthens rather than weakens the surrounding argument.
- Table 12's caption claimed its seven tasks were those of Mantri et al. (2026). They are from that paper's appendix Table 12, not its Table 3; only two of seven overlap.
- The RoBERTa `published_tau` arm is significant on both tests uncorrected on STS-B ($+0.32$, $t = 3.13$, $p = 0.035$) and was not marked.
- "Jumping-knowledge mode is uniform within a task" holds on CoLA and STS-B and fails on MRPC and RTE, where one arm each departs.
- Most consequentially, the 2×2 of Table 6 spliced two cells from one campaign into two from another. The factorial has since re-run the shared cell and disagrees with the original on all 15 seeds (Appendix D), which moves the "geometry alone" component of Stage C's gain from $+0.40$ to $+0.07$.

Everything else reproduced exactly: all 8 arms $\times$ 6 task-backbone columns of Table 4, every delta, t, sign-p and seed split in Tables 5 and 9, every row of Table 1, the re-measured densities of Table 8, the decoder block, the IMDB and MTEB values, and the learning-rate selection claims.

P LIMITATIONS

- **Stage C's gain is real but unattributed, and less attributable than we first reported.** The 2×2 of Table 6 decomposes it exactly, and none of the three components is significant at $n = 15$ even though the total is. Resolving the interaction needs about 65 seeds. Worse, the size of the geometry component depends on which draw of the shared cell is subtracted ($+0.40$ spliced, $+0.07$ within one campaign). Until the powered run lands we can say the effect requires both the denser graph and the curved message passing, but not which is responsible. It also does not reach MRPC and has never been run on a decoder.
- **A seed does not identify a run** (Appendix D). Within-campaign paired tests are unaffected, but pairing recovers much less variance than we claimed on CoLA, and no table may splice cells across campaigns. We did not discover this until the audit for this revision.
- **Our Stage C implementation duplicates self-loops** (Appendix N). The forward-pass effect is about 5% of a dense-to-sparse graph change, it differs between the two hyperbolic cells of the factorial, and it has not yet been corrected and re-run.

- **We select the epoch on the split we report**, because GLUE test labels are not public. This is a smaller version of the practice we criticise in §3. Splitting the development set into selection and reporting halves would remove the objection, and is the single most valuable improvement to this protocol.
- **One reproduction residual is unexplained.** STS-B stays 0.87 short under every estimator we tried; we tested and ruled out the leading hypothesis (Appendix K).
- **The scale factorial is underpowered.** At $n = 5$ the exact sign test cannot fall below $p = 0.0625$, so no contrast in Table 10 can meet our own bar.
- **Non-Euclidean layers may need structure-aware optimisation.** We gave every arm the full grid, and the search did move the learning rate per arm and per task rather than pinning it to a flat-tuned value. But we did not implement per-manifold Riemannian optimisation or curvature-aware initialisation. This is the strongest available objection to our Stage B negative, and we state it rather than merely acknowledging it.
- **Every task we ran is small, and that may explain Stage B.** CoLA, STS-B, MRPC and RTE have 8,551, 5,749, 3,668 and 2,490 training examples. Stage B hurts by amplifying seed variance, which more data suppresses, so its harm may not survive at scale — and RTE, the smallest, is the one task where B-containing arms trend positive. We could not test this: at 495 runs per task our design costs an estimated 170 hours for SST-2, 264 for QNLI, and 38 and 41 days for QQP and MNLI. A reduced design — four arms, confirmation seeds only, reusing the CoLA-selected configuration — brings SST-2 to roughly 21 GPU-hours and is the single most valuable experiment this paper is missing.
- **RTE is underpowered and contributes nothing.** With 277 evaluation examples and a seed standard deviation of 3.36, its minimum detectable effect is 2.06 accuracy points. All seven hyperbolic arms are nominally positive there and not one is significant. We include it for completeness and base no conclusion on it.
- **Coverage.** Mantri et al. (2026) report ten GLUE tasks on six backbones, plus IMDB and MTEB. We report four GLUE tasks with the full nine-arm design on BERT, two with the same design on RoBERTa, one on TinyLlama, a synthetic noise task, and IMDB plus seven MTEB tasks at a single seed for the graph stage only, the latter with an untrained pooler. We never ran Mean, Max, `[CLS]` or AdaPool on GLUE, so Table 4 reproduces the *layout* of the original's Table 1 but not its content; our rows are GLOT's own components.
- **The stages replicate; their interaction does not.** Stage B's harm reproduces across two encoders and a decoder at closely matching effect sizes, but the sign of the B×C interaction differs between all four encoder settings (§4.4), and at the extremes it is Bonferroni-significant in both directions. We have no account of why. This bounds what a component-wise study of this kind can deliver: individual stages generalise, but a two-stage combination has to be measured on the backbone and task where it will be used.